

# Public Health Sciences

*“It is a mathematical fact that fifty percent of all doctors graduate in the bottom half of their class.”*

—Author Unknown

*“There are two kinds of statistics: the kind you look up and the kind you make up.”*

—Rex Stout

*“On a long enough time line, the survival rate for everyone drops to zero.”*

—Chuck Palahniuk

*“There are three kinds of lies: lies, damned lies, and statistics.”*

—Mark Twain

A heterogenous mix of epidemiology, biostatistics, ethics, law, healthcare delivery, patient safety, quality improvement, and more falls under the heading of public health sciences. Biostatistics and epidemiology are the foundations of evidence-based medicine and are very high yield. Make sure you can apply biostatistical concepts such as sensitivity, specificity, and predictive values in a problem-solving format.

Medical ethics questions may seem less concrete than questions from other disciplines. For example, if a patient does or says something, what should you do or say in response? Many medical students do not diligently study these topics because the material is felt to be easy or a matter of common sense. In our opinion, this is a missed opportunity.

In addition, the key aspects of the doctor-patient relationship (eg, communication skills) are high yield. Last, the exam has also recently added an emphasis on patient safety and quality improvement topics, which are discussed in this chapter.

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## ► PUBLIC HEALTH SCIENCES—EPIDEMIOLOGY &amp; BIOSTATISTICS

## Observational studies

| STUDY TYPE                    | DESIGN                                                                                                                                                                                                                                                                                                                       | MEASURES/EXAMPLE                                                                                        |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Cross-sectional study</b>  | Frequency of disease and frequency of risk-related factors are assessed in the present.<br>Asks, “What is happening?”                                                                                                                                                                                                        | Disease prevalence.<br>Can show risk factor association with disease, but does not establish causality. |
| <b>Case-control study</b>     | Compares a group of people with disease to a group without disease.<br>Looks to see if odds of prior exposure or risk factor differs by disease state.<br>Asks, “What happened?”                                                                                                                                             | Odds ratio (OR).<br>Patients with COPD had higher odds of a history of smoking than those without COPD. |
| <b>Cohort study</b>           | Compares a group with a given exposure or risk factor to a group without such exposure.<br>Looks to see if exposure or risk factor is associated with later development of disease.<br>Can be prospective (asks, “Who will develop disease?”) or retrospective (asks, “Who developed the disease [exposed vs nonexposed]?”). | Relative risk (RR).<br>“Smokers had a higher risk of developing COPD than nonsmokers.”                  |
| <b>Twin concordance study</b> | Compares the frequency with which both monozygotic twins vs both dizygotic twins develop the same disease.                                                                                                                                                                                                                   | Measures heritability and influence of environmental factors (“nature vs nurture”).                     |
| <b>Adoption study</b>         | Compares siblings raised by biological vs adoptive parents.                                                                                                                                                                                                                                                                  | Measures heritability and influence of environmental factors.                                           |

## Clinical trial

Experimental study involving humans. Compares therapeutic benefits of 2 or more treatments, or of treatment and placebo. Study quality improves when study is randomized, controlled, and double-blinded (ie, neither patient nor doctor knows whether the patient is in the treatment or control group). Triple-blind refers to the additional blinding of the researchers analyzing the data. Four phases (“Does the drug **SWIM**?”).

| DRUG TRIALS      | TYPICAL STUDY SAMPLE                                                                                                                    | PURPOSE                                                                                                                |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Phase I</b>   | Small number of healthy volunteers.                                                                                                     | “Is it <b>S</b> afe?” Assesses safety, toxicity, pharmacokinetics, and pharmacodynamics.                               |
| <b>Phase II</b>  | Small number of patients with disease of interest.                                                                                      | “Does it <b>W</b> ork?” Assesses treatment efficacy, optimal dosing, and adverse effects.                              |
| <b>Phase III</b> | Large number of patients randomly assigned either to the treatment under investigation or to the best available treatment (or placebo). | “Is it as good or better?” Compares the new treatment to the current standard of care (any <b>I</b> mprovement?).      |
| <b>Phase IV</b>  | Postmarketing surveillance of patients after treatment is approved.                                                                     | “Can it stay?” Detects rare or long-term adverse effects. Can result in treatment being withdrawn from <b>M</b> arket. |

### Evaluation of diagnostic tests

Uses  $2 \times 2$  table comparing test results with the actual presence of disease. TP = true positive; FP = false positive; TN = true negative; FN = false negative.

Sensitivity and specificity are fixed properties of a test. PPV and NPV vary depending on disease prevalence in population being tested.

|      |   | Disease                           |                                   |                                                                                                                                       |
|------|---|-----------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
|      |   | ⊕                                 | ⊖                                 |                                                                                                                                       |
| Test | ⊕ | TP                                | FP                                | PPV<br>= $TP / (TP + FP)$                                                                                                             |
|      | ⊖ | FN                                | TN                                | NPV<br>= $TN / (TN + FN)$                                                                                                             |
|      |   | Sensitivity<br>= $TP / (TP + FN)$ | Specificity<br>= $TN / (TN + FP)$ | Prevalence<br>$\frac{TP + FN}{TP + FN + FP + TN}$  |

#### Sensitivity (true-positive rate)

Proportion of all people with disease who test positive, or the probability that when the disease is present, the test is positive.

Value approaching 100% is desirable for **ruling out** disease and indicates a **low false-negative rate**. High sensitivity test used for screening in diseases with low prevalence.

$$= TP / (TP + FN)$$

$$= 1 - \text{false-negative rate}$$

**SN-N-OUT** = highly **Se**nsitive test, when **N**egative, rules **OUT** disease

If sensitivity is 100%, then FN is zero. So, all negatives must be TNs.

#### Specificity (true-negative rate)

Proportion of all people without disease who test negative, or the probability that when the disease is absent, the test is negative.

Value approaching 100% is desirable for **ruling in** disease and indicates a **low false-positive rate**. High specificity test used for confirmation after a positive screening test.

$$= TN / (TN + FP)$$

$$= 1 - \text{false-positive rate}$$

**SP-P-IN** = highly **SP**ecific test, when **P**ositive, rules **IN** disease

If specificity is 100%, then FP is zero. So, all positives must be TPs.

#### Positive predictive value

Proportion of positive test results that are true positive.

Probability that a person who has a positive test result actually has the disease.

$$PPV = TP / (TP + FP)$$

PPV varies directly with pretest probability (baseline risk, such as prevalence of disease):  
high pretest probability → high PPV

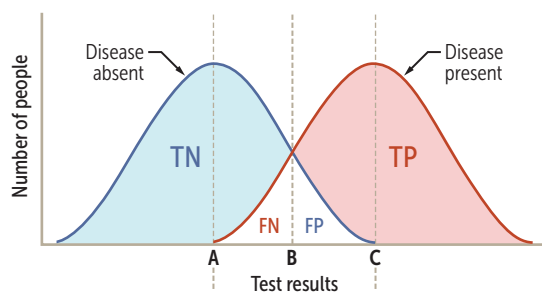
#### Negative predictive value

Proportion of negative test results that are true negative.

Probability that a person with a negative test result actually does not have the disease.

$$NPV = TN / (TN + FN)$$

NPV varies inversely with prevalence or pretest probability: high pretest probability → low NPV



#### POSSIBLE CUTOFF VALUES

- A = 100% sensitivity cutoff value
- B = practical compromise between specificity and sensitivity
- C = 100% specificity cutoff value

Lowering the cutoff point:  $\uparrow$  Sensitivity  $\uparrow$  NPV  
B  $\rightarrow$  A ( $\uparrow$  FP  $\downarrow$  FN)  $\downarrow$  Specificity  $\downarrow$  PPV

Raising the cutoff point:  $\uparrow$  Specificity  $\uparrow$  PPV  
B  $\rightarrow$  C ( $\uparrow$  FN  $\downarrow$  FP)  $\downarrow$  Sensitivity  $\downarrow$  NPV

For example, in diabetes screening, raising the blood glucose cutoff level at which a patient is diagnosed will  $\downarrow$  sensitivity,  $\uparrow$  specificity,  $\uparrow$  PPV, and  $\downarrow$  NPV. The opposite changes occur with decreasing the blood glucose cutoff level. 

### Likelihood ratio

LRs can be multiplied with pretest odds of disease to estimate posttest odds.  $LR^+ > 10$  and/or  $LR^- < 0.1$  are easy-to-remember indicators of a very useful diagnostic test.

$$LR^+ = \frac{\text{sensitivity}}{1 - \text{specificity}} = \frac{\text{True positive rate}}{\text{False positive rate}}$$

$$LR^- = \frac{1 - \text{sensitivity}}{\text{specificity}} = \frac{\text{False negative rate}}{\text{True negative rate}}$$

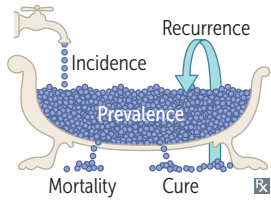
**Quantifying risk**

Definitions and formulas are based on the classic  $2 \times 2$  or contingency table.

|                                |         |   |
|--------------------------------|---------|---|
|                                | Disease |   |
|                                | ⊕       | ⊖ |
| Risk factor<br>or intervention | ⊕       | a |
|                                | ⊖       | b |
|                                | c       | d |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                        |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Odds ratio</b>              | Typically used in case-control studies. OR depicts the odds of an event (eg, disease) occurring giving a certain exposure (a/b) vs the odds of an event occurring in the absence of that exposure (c/d).                                                                                                                                                                                                                                                                                | $OR = \frac{a/b}{c/d} = \frac{ad}{bc}$ |
| <b>Relative risk</b>           | Typically used in cohort studies. Risk of developing disease in the exposed group divided by risk in the unexposed group (eg, if 21% of smokers develop lung cancer vs 1% of nonsmokers, $RR = 21/1 = 21$ ). For rare diseases (low prevalence), OR approximates RR.<br>$RR = 1 \rightarrow$ no association between exposure and disease.<br>$RR > 1 \rightarrow$ exposure associated with ↑ disease occurrence.<br>$RR < 1 \rightarrow$ exposure associated with ↓ disease occurrence. | $RR = \frac{a/(a+b)}{c/(c+d)}$         |
| <b>Attributable risk</b>       | The difference in risk between exposed and unexposed groups, or the proportion of disease occurrences that are attributable to the exposure (eg, if risk of lung cancer in smokers is 21% and risk in nonsmokers is 1%, then 20% of the lung cancer risk in smokers is attributable to smoking).                                                                                                                                                                                        | $AR = \frac{a}{a+b} - \frac{c}{c+d}$   |
| <b>Relative risk reduction</b> | The proportion of risk reduction attributable to the intervention as compared to a control (eg, if 2% of patients who receive a flu shot develop the flu, while 8% of unvaccinated patients develop the flu, then $RR = 2/8 = 0.25$ , and $RRR = 0.75$ ).                                                                                                                                                                                                                               | $RRR = 1 - RR$                         |
| <b>Absolute risk reduction</b> | The difference in risk (not the proportion) attributable to the intervention as compared to a control (eg, if 8% of people who receive a placebo vaccine develop the flu vs 2% of people who receive a flu vaccine, then $ARR = 8\% - 2\% = 6\% = .06$ ).                                                                                                                                                                                                                               | $ARR = \frac{c}{c+d} - \frac{a}{a+b}$  |
| <b>Number needed to treat</b>  | Number of patients who need to be treated for 1 patient to benefit. Lower number = better treatment.                                                                                                                                                                                                                                                                                                                                                                                    | $NNT = 1/ARR$                          |
| <b>Number needed to harm</b>   | Number of patients who need to be exposed to a risk factor for 1 patient to be harmed. Higher number = safer exposure.                                                                                                                                                                                                                                                                                                                                                                  | $NNH = 1/AR$                           |

### Incidence vs prevalence



Incidence =  $\frac{\text{\# of new cases}}{\text{\# of people at risk}}$  (during a specified time period)

Prevalence =  $\frac{\text{\# of existing cases}}{\text{Total \# of people in a population}}$  (at a point in time)

$\frac{\text{Prevalence}}{1 - \text{prevalence}} = \frac{\text{Incidence rate} \times \text{average duration of disease}}{1 - \text{prevalence}}$

Prevalence  $\approx$  incidence for short duration disease (eg, common cold).

Prevalence  $>$  incidence for chronic diseases, due to large # of existing cases (eg, diabetes).

**Incidence** looks at new cases (**incidents**).

Prevalence looks at **all** current cases.

Prevalence  $\sim$  pretest probability.

$\uparrow$  prevalence  $\rightarrow \uparrow$  PPV and  $\downarrow$  NPV.

### Precision vs accuracy

#### Precision (reliability)

The consistency and reproducibility of a test.  
The absence of random variation in a test.

Random error  $\downarrow$  precision in a test.

$\uparrow$  precision  $\rightarrow \downarrow$  standard deviation.

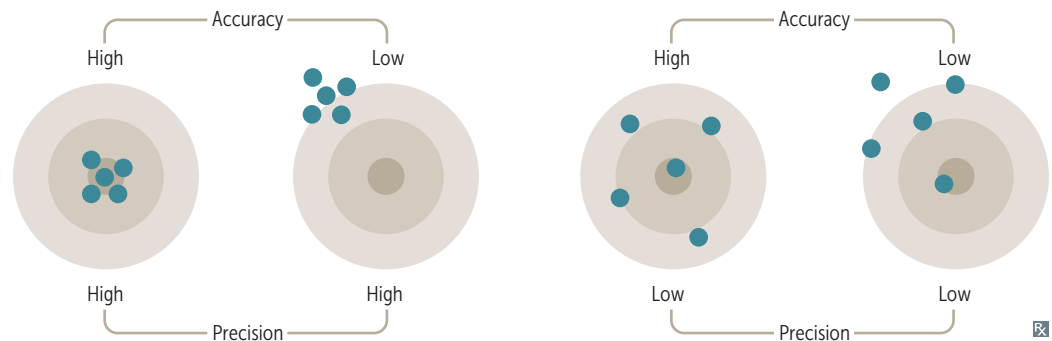
$\uparrow$  precision  $\rightarrow \uparrow$  statistical power ( $1 - \beta$ ).

#### Accuracy (validity)

The trueness of test measurements.

The absence of systematic error or bias in a test.

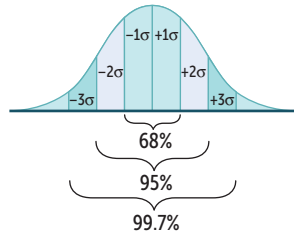
Systematic error  $\downarrow$  accuracy in a test.



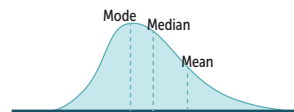
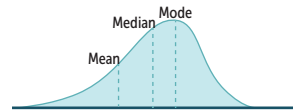
**Bias and study errors**

| TYPE                     | DEFINITION                                                                                                                                                                                                                                             | EXAMPLES                                                                                                                                                                                                                                                                  | STRATEGY TO REDUCE BIAS                                                                                                                                                                                      |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recruiting participants  |                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                              |
| Selection bias           | Nonrandom sampling or treatment allocation of subjects such that study population is not representative of target population (eg, study participants included based on adherence or other criteria related to outcome). Most commonly a sampling bias. | Berkson bias—study population selected from hospital is less healthy than general population<br>Healthy worker effect—study population is healthier than the general population<br>Non-response bias—participating subjects differ from nonrespondents in meaningful ways | Randomization<br>Ensure the choice of the right comparison/reference group                                                                                                                                   |
| Performing study         |                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                              |
| Recall bias              | Awareness of disorder alters recall by subjects; common in retrospective studies.                                                                                                                                                                      | Patients with disease recall exposure after learning of similar cases                                                                                                                                                                                                     | Decrease time from exposure to follow-up                                                                                                                                                                     |
| Measurement bias         | Information is gathered in a systemically distorted manner.                                                                                                                                                                                            | Association between HPV and cervical cancer not observed when using non-standardized classifications<br>Hawthorne effect—participants change their behavior in response to their awareness of being observed                                                              | Use objective, standardized, and previously tested methods of data collection that are planned ahead of time<br>Use placebo group                                                                            |
| Procedure bias           | Subjects in different groups are not treated the same.                                                                                                                                                                                                 | Patients in treatment group spend more time in highly specialized hospital units                                                                                                                                                                                          | Blinding and use of placebo reduce influence of participants and researchers on procedures and interpretation of outcomes as neither are aware of group allocation                                           |
| Observer-expectancy bias | Researcher’s belief in the efficacy of a treatment changes the outcome of that treatment (aka Pygmalion effect; self-fulfilling prophecy).                                                                                                             | If observer expects treatment group to show signs of recovery, then he is more likely to document positive outcomes                                                                                                                                                       |                                                                                                                                                                                                              |
| Interpreting results     |                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                              |
| Confounding bias         | When a factor is related to both the exposure and outcome, but not on the causal pathway → factor distorts or confuses effect of exposure on outcome.                                                                                                  | Pulmonary disease is more common in coal workers than the general population; however, people who work in coal mines also smoke more frequently than the general population                                                                                               | Multiple/repeated studies<br>Crossover studies (subjects act as their own controls)<br>Matching (patients with similar characteristics in both treatment and control groups)<br>Restriction<br>Randomization |
| Lead-time bias           | Early detection is confused with ↑ survival.                                                                                                                                                                                                           | Early detection makes it seem as though survival has increased, but the natural history of the disease has not changed                                                                                                                                                    | Measure “back-end” survival (adjust survival according to the severity of disease at the time of diagnosis)                                                                                                  |

**Statistical distribution**

|                                     |                                                                                                                                     |                                                                                                |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Measures of central tendency</b> | Mean = (sum of values)/(total number of values).                                                                                    | Most affected by outliers (extreme values).                                                    |
|                                     | Median = middle value of a list of data sorted from least to greatest.                                                              | If there is an even number of values, the median will be the average of the middle two values. |
|                                     | Mode = most common value.                                                                                                           | Least affected by outliers.                                                                    |
| <b>Measures of dispersion</b>       | Standard deviation = how much variability exists in a set of values, around the mean of these values.                               | $\sigma$ = SD; $n$ = sample size.                                                              |
|                                     | Standard error = an estimate of how much variability exists in a (theoretical) set of sample means around the true population mean. | Variance = $(SD)^2$ .<br>$SE = \sigma/\sqrt{n}$ .<br>$SE \downarrow$ as $n \uparrow$ .         |
| <b>Normal distribution</b>          | Gaussian, also called bell-shaped.<br>Mean = median = mode.                                                                         |             |

**Nonnormal distributions**

|                      |                                                                                                                                                          |                                                                                       |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Bimodal</b>       | Suggests two different populations (eg, metabolic polymorphism such as fast vs slow acetylators; age at onset of Hodgkin lymphoma; suicide rate by age). |   |
| <b>Positive skew</b> | Typically, mean > median > mode.<br>Asymmetry with longer tail on right.                                                                                 |  |
| <b>Negative skew</b> | Typically, mean < median < mode.<br>Asymmetry with longer tail on left.                                                                                  |  |

**Statistical hypotheses**

|                                       |                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                |  |         |  |  |       |       |                     |                         |                          |                             |                          |         |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------|--|--|-------|-------|---------------------|-------------------------|--------------------------|-----------------------------|--------------------------|---------|
| <b>Null (<math>H_0</math>)</b>        | Hypothesis of no difference or relationship (eg, there is no association between the disease and the risk factor in the population).     |                                                                                                                                                                                                                                                                                                                                                                                                |  |         |  |  |       |       |                     |                         |                          |                             |                          |         |
| <b>Alternative (<math>H_1</math>)</b> | Hypothesis of some difference or relationship (eg, there is some association between the disease and the risk factor in the population). | <table><tr><td></td><td colspan="2">Reality</td></tr><tr><td></td><td><math>H_1</math></td><td><math>H_0</math></td></tr><tr><td>Study rejects <math>H_0</math></td><td>Power<br/>(1 – <math>\beta</math>)</td><td><math>\alpha</math><br/>Type I error</td></tr><tr><td>Study does not reject <math>H_0</math></td><td><math>\beta</math><br/>Type II error</td><td>Correct</td></tr></table> |  | Reality |  |  | $H_1$ | $H_0$ | Study rejects $H_0$ | Power<br>(1 – $\beta$ ) | $\alpha$<br>Type I error | Study does not reject $H_0$ | $\beta$<br>Type II error | Correct |
|                                       | Reality                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                |  |         |  |  |       |       |                     |                         |                          |                             |                          |         |
|                                       | $H_1$                                                                                                                                    | $H_0$                                                                                                                                                                                                                                                                                                                                                                                          |  |         |  |  |       |       |                     |                         |                          |                             |                          |         |
| Study rejects $H_0$                   | Power<br>(1 – $\beta$ )                                                                                                                  | $\alpha$<br>Type I error                                                                                                                                                                                                                                                                                                                                                                       |  |         |  |  |       |       |                     |                         |                          |                             |                          |         |
| Study does not reject $H_0$           | $\beta$<br>Type II error                                                                                                                 | Correct                                                                                                                                                                                                                                                                                                                                                                                        |  |         |  |  |       |       |                     |                         |                          |                             |                          |         |

## Outcomes of statistical hypothesis testing

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Correct result</b>                     | <p>Stating that there is an effect or difference when one exists (null hypothesis rejected in favor of alternative hypothesis).</p> <p>Stating that there is not an effect or difference when none exists (null hypothesis not rejected).</p>                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Incorrect result</b>                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Type I error (<math>\alpha</math>)</b> | <p>Stating that there is an effect or difference when none exists (null hypothesis incorrectly rejected in favor of alternative hypothesis).</p> <p><math>\alpha</math> is the probability of making a type I error. <math>p</math> is judged against a preset <math>\alpha</math> level of significance (usually 0.05). If <math>p &lt; 0.05</math>, then there is less than a 5% chance that the data will show something that is not really there.</p>                                                                                                   | <p>Also known as false-positive error.</p> <p><math>\alpha</math> = you <b>a</b>ccused an innocent man.<br/>You can never “prove” the alternate hypothesis, but you can reject the null hypothesis as being very unlikely.</p>                                                                                                                                                                                                                      |
| <b>Type II error (<math>\beta</math>)</b> | <p>Stating that there is not an effect or difference when one exists (null hypothesis is not rejected when it is in fact false).</p> <p><math>\beta</math> is the probability of making a type II error. <math>\beta</math> is related to statistical power (<math>1 - \beta</math>), which is the probability of rejecting the null hypothesis when it is false.</p> <p>↑ power and ↓ <math>\beta</math> by:</p> <ul style="list-style-type: none"> <li>▪ ↑ sample size</li> <li>▪ ↑ expected effect size</li> <li>▪ ↑ precision of measurement</li> </ul> | <p>Also known as false-negative error.</p> <p><math>\beta</math> = you <b>b</b>lindly let the guilty man go free.<br/>If you ↑ sample size, you ↑ power. There is <b>power in numbers</b>.</p>                                                                                                                                                                                                                                                      |
| <b>Confidence interval</b>                | <p>Range of values within which the true mean of the population is expected to fall, with a specified probability.</p> <p>CI for population mean = <math>\bar{x} \pm Z(SE)</math></p> <p>The 95% CI (corresponding to <math>\alpha = .05</math>) is often used.</p> <p>For the 95% CI, <math>Z = 1.96</math>.</p> <p>For the 99% CI, <math>Z = 2.58</math>.</p>                                                                                                                                                                                             | <p>If the 95% CI for a mean difference between 2 variables includes 0, then there is no significant difference and <math>H_0</math> is not rejected.</p> <p>If the 95% CI for odds ratio or relative risk includes 1, <math>H_0</math> is not rejected.</p> <p>If the CIs between 2 groups do not overlap → statistically significant difference exists.</p> <p>If the CIs between 2 groups overlap → usually no significant difference exists.</p> |



## Common statistical tests

|                                         |                                                                                                                   |                                                                                                                                                          |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>t-test</b>                           | Checks differences between <b>means</b> of <b>2</b> groups.                                                       | Tea is <b>meant</b> for <b>2</b> .<br>Example: comparing the mean blood pressure between men and women.                                                  |
| <b>ANOVA</b>                            | Checks differences between means of <b>3</b> or more groups.                                                      | <b>3</b> words: <b>AN</b> alysis <b>Of</b> <b>VA</b> riance.<br>Example: comparing the mean blood pressure between members of 3 different ethnic groups. |
| <b>Chi-square (<math>\chi^2</math>)</b> | Checks differences between 2 or more percentages or proportions of <b>categorical</b> outcomes (not mean values). | Pronounce <b>Chi-tegorical</b> .<br>Example: comparing the percentage of members of 3 different ethnic groups who have essential hypertension.           |

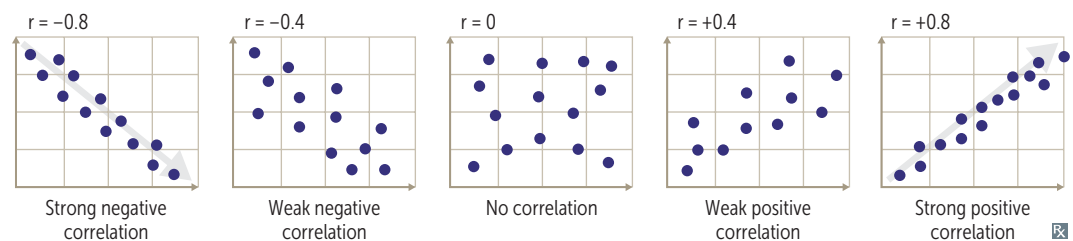
## Pearson correlation coefficient

$r$  is always between  $-1$  and  $+1$ . The closer the absolute value of  $r$  is to 1, the stronger the linear correlation between the 2 variables.

Positive  $r$  value  $\rightarrow$  positive correlation (as one variable  $\uparrow$ , the other variable  $\uparrow$ ).

Negative  $r$  value  $\rightarrow$  negative correlation (as one variable  $\uparrow$ , the other variable  $\downarrow$ ).

Coefficient of determination =  $r^2$  (amount of variance in one variable that can be explained by variance in another variable).



## ► BEHAVIORAL SCIENCE—ETHICS

## Core ethical principles

|                       |                                                                                                                                                                                                                                                                          |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Autonomy</b>       | Obligation to respect patients as individuals (truth-telling, confidentiality), to create conditions necessary for autonomous choice (informed consent), and to honor their preference in accepting or not accepting medical care.                                       |
| <b>Beneficence</b>    | Physicians have a special ethical (fiduciary) duty to act in the patient's best interest. May conflict with autonomy (an informed patient has the right to decide) or what is best for society (eg, mandatory TB treatment). Traditionally, patient interest supersedes. |
| <b>Nonmaleficence</b> | "Do no harm." Must be balanced against beneficence; if the benefits outweigh the risks, a patient may make an informed decision to proceed (most surgeries and medications fall into this category).                                                                     |
| <b>Justice</b>        | To treat persons fairly and equitably. This does not always imply equally (eg, triage).                                                                                                                                                                                  |

**Informed consent**

A process (not just a document/signature) that requires:

- Disclosure: discussion of pertinent information
- Understanding: ability to comprehend
- Capacity: ability to reason and make one's own decisions (distinct from competence, a legal determination)
- Voluntariness: freedom from coercion and manipulation

Patients must have an intelligent understanding of their diagnosis and the risks/benefits of proposed treatment and alternative options, including no treatment.

Patient must be informed that he or she can revoke written consent at any time, even orally.

Exceptions to informed consent:

- Patient lacks decision-making capacity or is legally incompetent
- Implied consent in an emergency
- Therapeutic privilege—withholding information when disclosure would severely harm the patient or undermine informed decision-making capacity
- Waiver—patient explicitly waives the right of informed consent

**Consent for minors**

A minor is generally any person < 18 years old. Parental consent laws in relation to healthcare vary by state. In general, parental consent should be obtained, but exceptions exist for emergency treatment (eg, blood transfusions) or if minor is legally emancipated (eg, married, self supporting, or in the military).

Situations in which parental consent is usually not required:

- **Sex** (contraception, STIs, pregnancy)
- **Drugs** (substance abuse)
- **Rock and roll** (emergency/trauma)

Physicians should always encourage healthy minor-guardian communication.

Physician should seek a minor's assent even if their consent is not required.

**Decision-making capacity**

Physician must determine whether the patient is psychologically and legally capable of making a particular healthcare decision. Note that decisions made with capacity cannot be revoked simply if the patient later loses capacity.

Capacity is determined by a physician for a specific healthcare-related decision (eg, to refuse medical care). Competency is determined by a judge and usually refers to more global categories of decision making (eg, legally unable to make any healthcare-related decision).

Components:

- Patient is ≥ 18 years old or otherwise legally emancipated
- Patient makes and communicates a choice
- Patient is informed (knows and understands)
- Decision remains stable over time
- Decision is consistent with patient's values and goals, not clouded by a mood disorder
- Decision is not a result of altered mental status (eg, delirium, psychosis, intoxication)

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advance directives</b>        | Instructions given by a patient in anticipation of the need for a medical decision. Details vary per state law.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Oral advance directive</b>    | Incapacitated patient's prior oral statements commonly used as guide. Problems arise from variance in interpretation. If patient was informed, directive was specific, patient made a choice, and decision was repeated over time to multiple people, then the oral directive is more valid.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Written advance directive</b> | Specifies specific healthcare interventions that a patient anticipates he or she would accept or reject during treatment for a critical or life-threatening illness. A living will is an example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Medical power of attorney</b> | Patient designates an agent to make medical decisions in the event that he/she loses decision-making capacity. Patient may also specify decisions in clinical situations. Can be revoked by patient if decision-making capacity is intact. More flexible than a living will.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Do not resuscitate order</b>  | DNR order applies to cardiopulmonary resuscitation (CPR). Other resuscitative measures that follow (cardioversion, intubation) are also typically avoided.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Surrogate decision-maker</b>  | If a patient loses decision-making capacity and has not prepared an advance directive, individuals (surrogates) who know the patient must determine what the patient would have done. Priority of surrogates: <b>spouse</b> → adult <b>Children</b> → <b>Parents</b> → <b>Siblings</b> → other relatives (the <b>spouse ChiPS</b> in).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Confidentiality</b>           | <p>Confidentiality respects patient privacy and autonomy. If patient is not present or is incapacitated, disclosing information to family and friends should be guided by professional judgment of patient's best interest. The patient may voluntarily waive the right to confidentiality (eg, insurance company request).</p> <p>General principles for exceptions to confidentiality:</p> <ul style="list-style-type: none"> <li>▪ Potential physical harm to others is serious and imminent</li> <li>▪ Likelihood of harm to self is great</li> <li>▪ No alternative means exist to warn or to protect those at risk</li> <li>▪ Physicians can take steps to prevent harm</li> </ul> <p>Examples of exceptions to patient confidentiality (many are state-specific) include the following ("The physician's good judgment <b>SAVED</b> the day"):</p> <ul style="list-style-type: none"> <li>▪ <b>Suicidal/homicidal</b> patients</li> <li>▪ <b>Abuse</b> (children, elderly, and/or prisoners)</li> <li>▪ Duty to protect—State-specific laws that sometimes allow physician to inform or somehow protect potential <b>Victim</b> from harm.</li> <li>▪ <b>Epileptic</b> patients and other impaired automobile drivers.</li> <li>▪ Reportable <b>Diseases</b> (eg, STIs, hepatitis, food poisoning); physicians may have a duty to warn public officials, who will then notify people at risk. Dangerous communicable diseases, such as TB or Ebola, may require involuntary treatment.</li> </ul> |

**Ethical situations**

| SITUATION                                                                                                                                            | APPROPRIATE RESPONSE                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Patient is not adherent.                                                                                                                             | Attempt to identify the reason for nonadherence and determine his/her willingness to change; do not coerce the patient into adhering and do not refer him/her to another physician.                                                                                                                                                                                                                                                                               |
| Patient desires an unnecessary procedure.                                                                                                            | Attempt to understand why the patient wants the procedure and address underlying concerns. Do not refuse to see the patient and do not refer him/her to another physician. Avoid performing unnecessary procedures.                                                                                                                                                                                                                                               |
| Patient has difficulty taking medications.                                                                                                           | Provide written instructions; attempt to simplify treatment regimens; use teach-back method (ask patient to repeat regimen back to physician) to ensure comprehension.                                                                                                                                                                                                                                                                                            |
| Family members ask for information about patient's prognosis.                                                                                        | Avoid discussing issues with relatives without the patient's permission.                                                                                                                                                                                                                                                                                                                                                                                          |
| A patient's family member asks you not to disclose the results of a test if the prognosis is poor because the patient will be "unable to handle it." | Attempt to identify why the family member believes such information would be detrimental to the patient's condition. Explain that as long as the patient has decision-making capacity and does not indicate otherwise, communication of information concerning his/her care will not be withheld. However, if you believe the patient might seriously harm himself or others if informed, then you may invoke therapeutic privilege and withhold the information. |
| A 17-year-old girl is pregnant and requests an abortion.                                                                                             | Many states require parental notification or consent for minors for an abortion. Unless there are specific medical risks associated with pregnancy, a physician should not sway the patient's decision for an elective abortion (regardless of maternal age or fetal condition).                                                                                                                                                                                  |
| A 15-year-old girl is pregnant and wants to keep the child. Her parents want you to tell her to give the child up for adoption.                      | The patient retains the right to make decisions regarding her child, even if her parents disagree. Provide information to the teenager about the practical issues of caring for a baby. Discuss the options, if requested. Encourage discussion between the teenager and her parents to reach the best decision.                                                                                                                                                  |
| A terminally ill patient requests physician assistance in ending his/her own life.                                                                   | In the overwhelming majority of states, refuse involvement in any form of physician-assisted suicide. Physicians may, however, prescribe medically appropriate analgesics that coincidentally shorten the patient's life.                                                                                                                                                                                                                                         |
| Patient is suicidal.                                                                                                                                 | Assess the seriousness of the threat. If it is serious, suggest that the patient remain in the hospital voluntarily; patient can be hospitalized involuntarily if he/she refuses.                                                                                                                                                                                                                                                                                 |
| Patient states that he/she finds you attractive.                                                                                                     | Ask direct, closed-ended questions and use a chaperone if necessary. Romantic relationships with patients are never appropriate.                                                                                                                                                                                                                                                                                                                                  |
| A woman who had a mastectomy says she now feels "ugly."                                                                                              | Find out why the patient feels this way. Do not offer falsely reassuring statements (eg, "You still look good").                                                                                                                                                                                                                                                                                                                                                  |
| Patient is angry about the long time he/she spent in the waiting room.                                                                               | Acknowledge the patient's anger, but do not take a patient's anger personally. Apologize for any inconvenience. Stay away from efforts to explain the delay.                                                                                                                                                                                                                                                                                                      |
| Patient is upset with the way he/she was treated by another doctor.                                                                                  | Suggest that the patient speak directly to that physician regarding his/her concerns. If the problem is with a member of the office staff, tell the patient you will speak to that person.                                                                                                                                                                                                                                                                        |
| An invasive test is performed on the wrong patient.                                                                                                  | Regardless of the outcome, a physician is ethically obligated to inform a patient that a mistake has been made.                                                                                                                                                                                                                                                                                                                                                   |

**Ethical situations (continued)**

| SITUATION                                                                                                                                                                 | APPROPRIATE RESPONSE                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A patient requires a treatment not covered by his/her insurance.                                                                                                          | Never limit or deny care because of the expense in time or money. Discuss all treatment options with patients, even if some are not covered by their insurance companies.                                                                                                                                                                                                                                                                   |
| A 7-year-old boy loses a sister to cancer and now feels responsible.                                                                                                      | At ages 5–7, children begin to understand that death is permanent, that all life functions end completely at death, and that everything that is alive eventually dies. Provide a direct, concrete description of his sister's death. Avoid clichés and euphemisms. Reassure that the boy is not responsible. Identify and normalize fears and feelings. Encourage play and healthy coping behaviors (eg, remembering her in his own way).   |
| Patient is victim of intimate partner violence.                                                                                                                           | Ask if patient is safe and has an emergency plan. Do not necessarily pressure patient to leave his or her partner, or disclose the incident to the authorities (unless required by state law).                                                                                                                                                                                                                                              |
| Patient wants to try alternative or holistic medicine.                                                                                                                    | Find out why and allow patient to do so as long as there are no contraindications, medication interactions, or adverse effects to the new treatment.                                                                                                                                                                                                                                                                                        |
| Physician colleague presents to work impaired.                                                                                                                            | If impaired or incompetent, colleague is a threat to patient safety. Report the situation to local supervisory personnel. Should the organization fail to take action, alert the state licensing board.                                                                                                                                                                                                                                     |
| Patient is officially determined to suffer brain death. Patient's family insists on maintaining life support indefinitely because patient is still moving when touched.   | Gently explain to family that there is no chance of recovery, and that brain death is equivalent to death. Movement is due to spinal arc reflex and is not voluntary. Bring case to appropriate ethics board regarding futility of care and withdrawal of life support.                                                                                                                                                                     |
| A pharmaceutical company offers you a sponsorship in exchange for advertising its new drug.                                                                               | Reject this offer. Generally, decline gifts and sponsorships to avoid any appearance of conflict of interest. The AMA Code of Ethics does make exceptions for gifts directly benefitting patients; gifts of minimal value; special funding for medical education of students, residents, fellows; grants whose recipients are chosen by independent institutional criteria; and funds that are distributed without attribution to sponsors. |
| An adult refuses care because it is against his/her religious beliefs.                                                                                                    | Work with the patient by either explaining the treatment or pursuing alternative treatments. However, a physician should never force a competent adult to receive care if it is contrary to the patient's religious beliefs.                                                                                                                                                                                                                |
| Mother and 15-year-old daughter are unresponsive following a car accident and are bleeding internally. Father says do not transfuse because they are Jehovah's Witnesses. | Transfuse daughter, but do not transfuse mother. Emergent care can be refused by the healthcare proxy for an adult, particularly when patient preferences are known or reasonably inferred, but not for a minor.                                                                                                                                                                                                                            |

## ► PUBLIC HEALTH SCIENCES—THE WELL PATIENT

**Early developmental milestones**

Milestone dates are ranges that have been approximated and vary by source. Children not meeting milestones may need assessment for potential developmental delay.

| AGE       | MOTOR                                                                                                                                                                                                                                                                                                                                                                           | SOCIAL                                                                                                                                                                                 | VERBAL/COGNITIVE                                                                                                                                                                  |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Infant    | Parents                                                                                                                                                                                                                                                                                                                                                                         | Start                                                                                                                                                                                  | Observing,                                                                                                                                                                        |
| 0–12 mo   | <b>P</b> rimitive reflexes disappear—Moro (by 3 mo), rooting (by 4 mo), palmar (by 6 mo), Babinski (by 12 mo)<br><b>P</b> osture—lifts head up prone (by 1 mo), rolls and sits (by 6 mo), crawls (by 8 mo), stands (by 10 mo), walks (by 12–18 mo)<br><b>P</b> icks—passes toys hand to hand (by 6 mo), <b>P</b> incer grasp (by 10 mo)<br><b>P</b> oints to objects (by 12 mo) | <b>S</b> ocial smile (by 2 mo)<br><b>S</b> tranger anxiety (by 6 mo)<br><b>S</b> eparation anxiety (by 9 mo)                                                                           | <b>O</b> rients—first to voice (by 4 mo), then to name and gestures (by 9 mo)<br><b>O</b> bject permanence (by 9 mo)<br><b>O</b> ratory—says “mama” and “dada” (by 10 mo)         |
| Toddler   | Child                                                                                                                                                                                                                                                                                                                                                                           | Rearing                                                                                                                                                                                | Working,                                                                                                                                                                          |
| 12–36 mo  | <b>C</b> ruises, takes first steps (by 12 mo)<br><b>C</b> limbs stairs (by 18 mo)<br><b>C</b> ubes stacked—number = age (yr) × 3<br><b>C</b> utlery—feeds self with fork and spoon (by 20 mo)<br><b>K</b> icks ball (by 24 mo)                                                                                                                                                  | <b>R</b> ecreation—parallel play (by 24–36 mo)<br><b>R</b> approchement—moves away from and returns to mother (by 24 mo)<br><b>R</b> ealization—core gender identity formed (by 36 mo) | <b>W</b> ords— <b>200</b> words by age <b>2</b> ( <b>2 zeros</b> ), <b>2</b> -word sentences                                                                                      |
| Preschool | Don't                                                                                                                                                                                                                                                                                                                                                                           | Forget, they're still                                                                                                                                                                  | Learning!                                                                                                                                                                         |
| 3–5 yr    | <b>D</b> rive—tricycle ( <b>3</b> wheels at <b>3</b> yr)<br><b>D</b> rawings—copies line or circle, stick figure (by 4 yr)<br><b>D</b> exterity—hops on one foot (by 4 yr), uses buttons or zippers, grooms self (by 5 yr)                                                                                                                                                      | <b>F</b> reedom—comfortably spends part of day away from mother (by 3 yr)<br><b>F</b> riends—cooperative play, has imaginary friends (by 4 yr)                                         | <b>L</b> anguage— <b>1000</b> words by age <b>3</b> ( <b>3 zeros</b> ), uses complete sentences and prepositions (by 4 yr)<br><b>L</b> egends—can tell detailed stories (by 4 yr) |

**Car seats for children**

Children should ride in rear-facing car seats until they are 2 years old and in car seats with a harness until they are 4 years. Older children should use a booster seat until they are 8 years old or until the seat belt fits properly. Children < 12 years old should not ride in a seat with a front-facing airbag.

**Changes in the elderly**

Sexual changes:

- Men—slower erection/ejaculation, longer refractory period.
- Women—vaginal shortening, thinning, and dryness.

Sleep patterns: ↓ REM and slow-wave sleep; ↑ sleep onset latency; ↑ early awakenings.

↑ suicide rate.

↓ vision and hearing.

↓ immune response.

↓ renal, pulmonary, and GI function.

↓ muscle mass, ↑ fat.

Intelligence does not decrease.

## ► PUBLIC HEALTH SCIENCES—HEALTHCARE DELIVERY

**Disease prevention**

|                  |                                                                                                            |                                                                                                                                                                                                                                |
|------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary</b>   | Prevent disease before it occurs (eg, HPV vaccination)                                                     |                                                                                                                                                                                                                                |
| <b>Secondary</b> | Screen early for and manage existing but asymptomatic disease (eg, Pap smear for cervical cancer)          |                                                                                                                                                                                                                                |
| <b>Tertiary</b>  | Treatment to reduce complications from disease that is ongoing or has long-term effects (eg, chemotherapy) | <b>Quaternary</b> —identifying patients at risk of unnecessary treatment, protecting from the harm of new interventions (eg, electronic sharing of patient records to avoid duplicating recent laboratory and imaging studies) |

**Major medical insurance plans**

|                                        |                                                                                                                                                                                                                                                                              |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Health Maintenance Organization</b> | Patients are restricted (except in emergencies) to a limited panel of providers who are in the network.<br>Payment is denied for any service that does not meet established, evidence-based guidelines.<br>Requires referral from primary care provider to see a specialist. |
| <b>Point of Service</b>                | Patients are allowed to see providers outside of the network, but have higher out-of-pocket costs, including higher copays and deductibles, for out-of-network services.<br>Requires referral from primary care provider to see a specialist.                                |
| <b>Preferred Provider Organization</b> | Patients are allowed to see physicians who are within or outside of the network. All services have higher copays and deductibles.<br>Does not require referral from primary care provider to see a specialist.                                                               |
| <b>Exclusive Provider Organization</b> | Patients are limited (except in emergencies) to a network of doctors, specialists, and hospitals.<br>Does not require referral from primary care provider to see a specialist.                                                                                               |

**Healthcare payment models**

|                                   |                                                                                                                                                                                                                                       |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Capitation</b>                 | Physicians receive a set amount per patient assigned to them per period of time, regardless of how much the patient uses the healthcare system.                                                                                       |
| <b>Discounted fee-for-service</b> | Patient pays for each individual service at a predetermined, discounted rate.                                                                                                                                                         |
| <b>Global payment</b>             | Patient pays for all expenses associated with a single incident of care with a single payment. Most commonly used during elective surgeries, as it covers the cost of surgery as well as the necessary pre- and postoperative visits. |

**Medicare and Medicaid**

Medicare and Medicaid—federal social healthcare programs that originated from amendments to the Social Security Act.

Medicare is available to patients  $\geq 65$  years old,  $< 65$  with certain disabilities, and those with end-stage renal disease.

Medicaid is joint federal and state health assistance for people with limited income and/or resources.

Medicar**E** is for **E**lderly.  
 Medicaid**D** is for **D**estitute.

The 4 parts of Medicare:

- Part **A**: Hospit**A**l insurance, home hospice care
- Part **B**: **B**asic medical **b**ills (eg, doctor's fees, diagnostic testing)
- Part **C**: (parts A + B = **C**ombo) delivered by approved private **c**ompanies
- Part **D**: Prescription **D**rugs

**Hospice care**

Medical care focused on providing comfort and palliation instead of definitive cure. Available to patients on Medicare or Medicaid and in most private insurance plans whose life expectancy is  $< 6$  months.

During end-of-life care, priority is given to improving the patient's comfort and relieving pain, and care often includes opioid medications. Facilitating comfort is prioritized over potential side effects (eg, respiratory depression). This prioritization of positive effects over negative effects is known as the **principle of double effect**.

**Common causes of death (US) by age**

|           | $< 1$ YR                        | 1–14 YR                  | 15–34 YR             | 35–44 YR             | 45–64 YR             | 65+ YR                      |
|-----------|---------------------------------|--------------------------|----------------------|----------------------|----------------------|-----------------------------|
| <b>#1</b> | Congenital malformations        | Unintentional injury     | Unintentional injury | Unintentional injury | Cancer               | Heart disease               |
| <b>#2</b> | Preterm birth                   | Cancer                   | Suicide              | Cancer               | Heart disease        | Cancer                      |
| <b>#3</b> | Maternal pregnancy complication | Congenital malformations | Homicide             | Heart disease        | Unintentional injury | Chronic respiratory disease |



**Hospitalized conditions with frequent readmissions**

|    | MEDICARE      | MEDICAID                             | PRIVATE INSURANCE                                    | UNINSURED                            |
|----|---------------|--------------------------------------|------------------------------------------------------|--------------------------------------|
| #1 | Congestive HF | Mood disorders                       | Maintenance of chemotherapy or radiotherapy          | Mood disorders                       |
| #2 | Septicemia    | Schizophrenia/psychotic disorders    | Mood disorders                                       | Alcohol-related disorders            |
| #3 | Pneumonia     | Diabetes mellitus with complications | Complications of surgical procedures or medical care | Diabetes mellitus with complications |

Readmission for any reason within 30 days of original admission.

## ► PUBLIC HEALTH SCIENCES—QUALITY AND SAFETY

**Safety culture**

Organizational environment in which everyone can freely bring up safety concerns without fear of censure. Facilitates error identification.

Event reporting systems collect data on errors for internal and external monitoring.

**Human factors design**

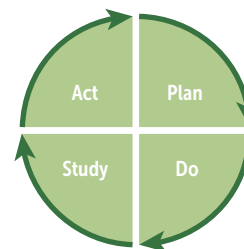
Forcing functions (those that prevent undesirable actions [eg, connecting feeding syringe to IV tubing]) are the most effective. Standardization improves process reliability (eg, clinical pathways, guidelines, checklists). Simplification reduces wasteful activities (eg, consolidating electronic medical records [EMRs]).

Deficient designs hinder workflow and lead to staff workarounds that bypass safety features (eg, patient ID barcodes affixed to computers due to unreadable wristbands).

**PDSA cycle**

Process improvement model to test changes in real clinical setting. Impact on patients:

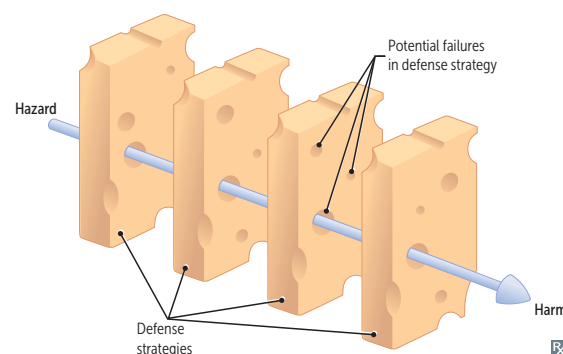
- **P**lan—define problem and solution
- **D**o—test new process
- **S**tudy—measure and analyze data
- **A**ct—integrate new process into regular workflow

**Quality measurements** Plotted on run and control charts.

|                  | MEASURE                          | EXAMPLE                                                                                 |
|------------------|----------------------------------|-----------------------------------------------------------------------------------------|
| <b>Outcome</b>   | Impact on patients               | Average HbA <sub>1c</sub> of patients with diabetes                                     |
| <b>Process</b>   | Performance of system as planned | Percentage of target patients whose HbA <sub>1c</sub> was measured in the past 6 months |
| <b>Balancing</b> | Impact on other systems/outcomes | Incidence of hypoglycemia among those patients                                          |

**Swiss cheese model**

Focuses on systems and conditions rather than an individual's error. The risk of a threat becoming a reality is mitigated by differing layers and types of defenses. Patient harm can occur despite multiple safeguards when “the holes in the cheese line up.”

**Types of medical errors**

May involve patient identification, diagnosis, monitoring, nosocomial infection, medications, procedures, devices, documentation, handoffs. Errors causing harmful outcomes must be disclosed to patients.

**Active error**

Occurs at level of frontline operator (eg, wrong IV pump dose programmed).

Immediate impact.

**Latent error**

Occurs in processes indirect from operator but impacts patient care (eg, different types of IV pumps used within same hospital).

Accident waiting to happen.

**Medical error analysis****Root cause analysis**

Uses records and participant interviews to identify all the underlying problems that led to an error. Categories of causes include process, people (providers or patients), environment, equipment, materials, management.

**R**etrospective approach applied after failure event to prevent recurrence. Plotted on fishbone (Ishikawa, cause-and-effect) diagram. Fix causes with corrective action plan.

**Failure mode and effects analysis**

Uses inductive reasoning to identify all the ways a process might fail and prioritize these by their probability of occurrence and impact on patients.

**F**orward-looking approach applied before process implementation to prevent failure occurrence.